Institute of Technology of Cambodia

Department of Applied Mathematics and Statistics

Group Project 3

Course Subject: Probability

Topic: Spam email probability

Probability Report (2025-2026)

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I. INTRODUCTION

1.1 Problem Statement

This project studies the probability of receiving spam emails. Spam emails are unwanted messages that are sent to many users through email systems. These emails usually contain advertisements, fake promotions, scams, or harmful links. Spam emails can waste users' time, fill up inbox storage, and make it difficult to find important messages. In some cases, spam emails may also contain phishing links that try to steal personal information such as passwords, bank account details, or other private data.

Nowadays, email is one of the most common communication methods used in schools, businesses, and daily life. Because of this, the number of spam emails continues to increase every day. Therefore, it is important to study spam emails and understand how they affect users and communication systems.

1.2 Importance of the problem in field

Probability helps analyze this problem by using mathematical methods to measure and predict the occurrence of spam emails. By using concepts such as probability distribution, expected value, variance, and standard deviation, we can understand the average number of spam emails received and how much the number changes over time. These results help identify spam email patterns and provide useful information about the behavior and consistency of spam email activity.

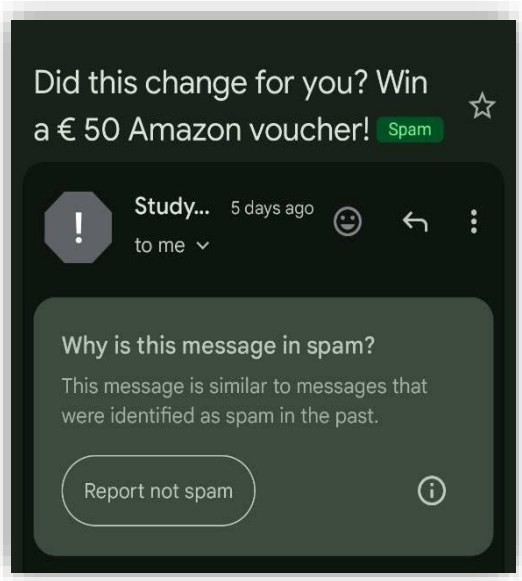
1.3 Role of probability in this study

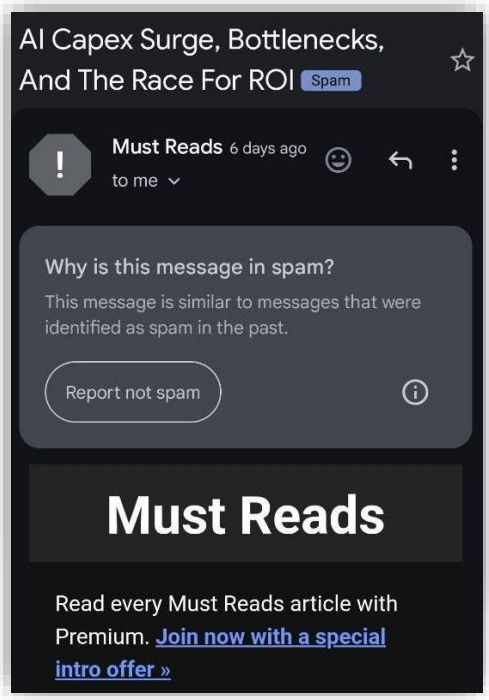
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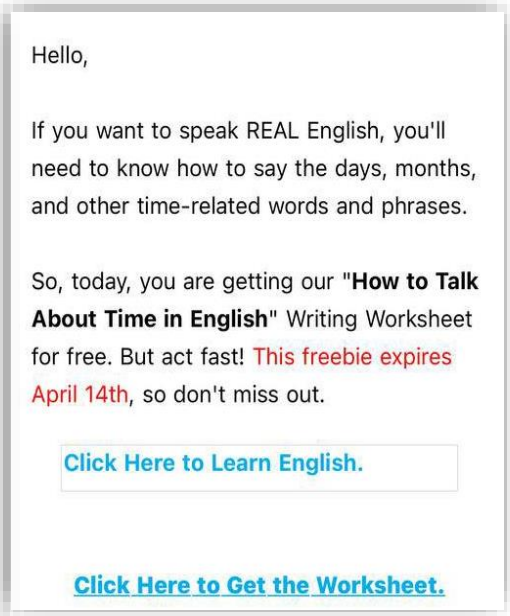
II. DESCRIPTION OF THE EXPERIMENT

2.1 Method of Data Collection

The data for this project was collected through a survey conducted among 30 students at ITC. Students were asked whether they had received spam emails in their email accounts. Those who had spam emails were asked to open their Gmail accounts and provide screenshots of the spam messages as evidence. From the survey, 9 students reported receiving spam emails from different categories such as Lottery Money, Education, AI/GitHub, Game, Storage Image, and promotional messages. The screenshots were recorded and analyzed to identify the number of spam categories received by each student. The collected data was then used for probability calculations.







2.2 Define the Random Variable

From the collected data, the random variable is defined as:

X = the number of spam categories received by an affected student during the two-weeks survey.

The survey was conducted among 30 students and 9 students were found to have received spam emails. Some students received spam from only one category, while others received spam from multiple categories.

So, the possible values of the random variable are:

$$X = \{1, 2, 3, 4\}$$

Where:

- X = 1 means a student receives spam from one category only
- X = 2 means a student received spam from two categories
- X = 3 means a student received spam from three categories
- X = 4 means a student received spam from categories

This random variable is a **discrete random variable** because the number of spam categories can only be counted using whole numbers.

2.3 Probability Distribution

Base on data receives: we have 30 students and we found 9 students who have the different and the same spam emails (students might have 1-2 or 3 spam emails).

Number of Student	Spam Category
Student 1	Lottery Money, Game, Storage Image
Student 2	Education
Student 3	AI / GitHub, Lottery Money, Education
Student 4	Education
Student 5	Education
Student 6	AI / GitHub
Student 7	Lottery Money, Storage Image
Student 8	Lottery Money
Student 9	AI / GitHub

Based on the collected data, the probability distribution table for the number of spam categories received by an affected student is shown below.

Probability Distribution Table

Number of Spam Categories X	Frequency	Probability P(X)
1	6	$\frac{6}{9} = 0.667$
2	1	$\frac{1}{9} = 0.111$
3	2	$\frac{2}{9} = 0.222$
4	0	$\frac{0}{9} = 0.000$

Check:

$$0.667 + 0.111 + 0.222 + 0.000 = 1.000$$

So, the total probability equals 1.000 which confirms that the probability distribution is valid.

Explanation of the Table:

The table shows that:

- Most affected students received spam from only one category.
- A smaller number of students received spam from two or three categories.
- No student received spam from four categories during the survey period.

The probability of a student receiving spam from only one category is the highest:

$$P(X = 1) = 0.667$$

This means there is about a 66.7% chance that an affected students receives spam from only one category.

The probability of receiving spam from three categories is:

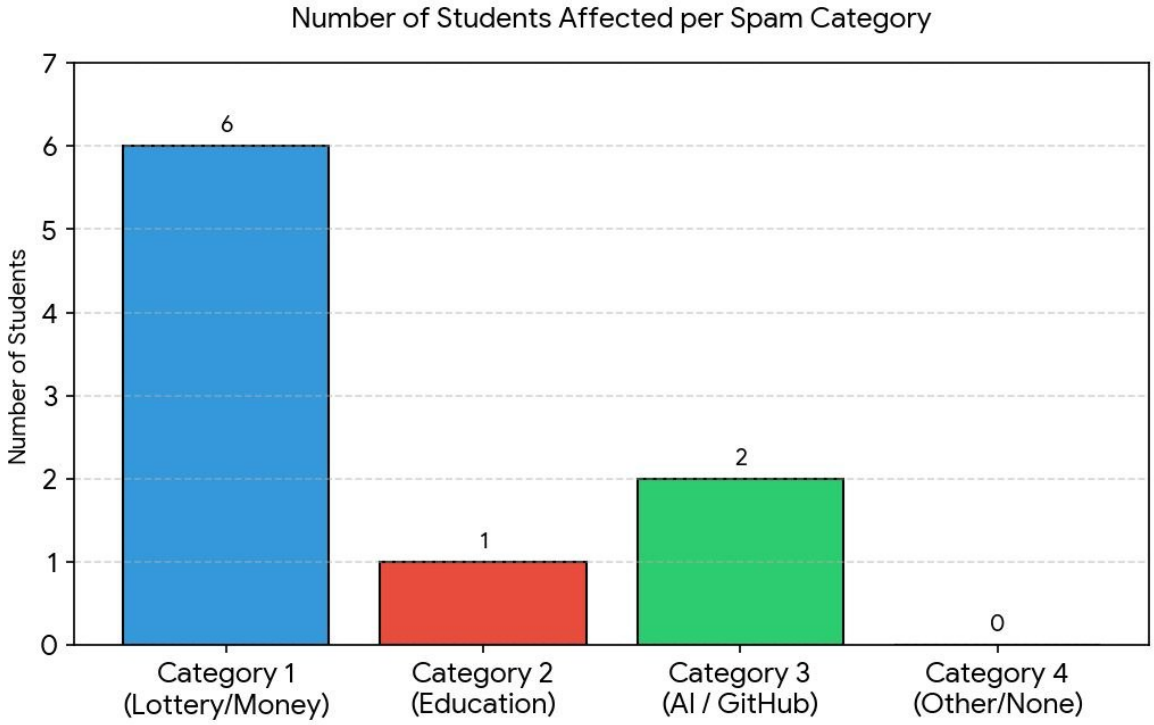
$$P(X = 3) = 0.222$$

This means there is a 22.2% chance that an affected student receives spam from three categories.

2.4 Probability Calculations

Base on data we identity the number of students affect by category spam:

Spam category (x)	Number of Students affect by spam
1	6
2	1
3	2
4	0
Total	9 students

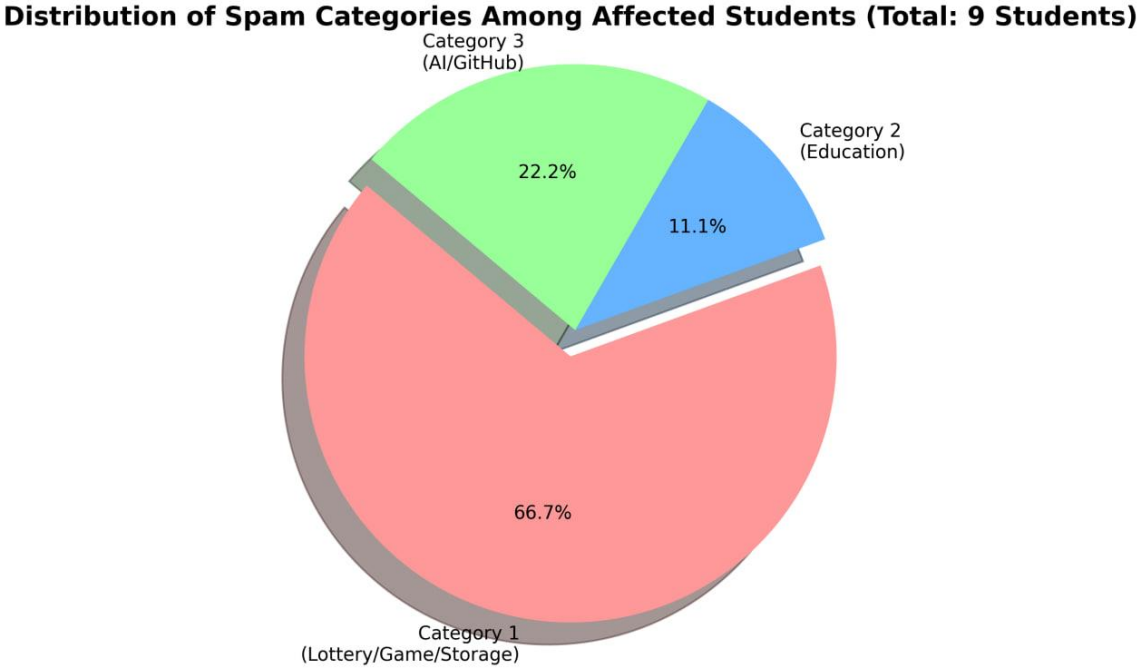


This chart highlights the exact count of students affected by each spam type.

Find value of each probabitity

$$P(X = x) = \frac{\text{Number of Students affect by spam}}{\text{Total students affects by spam}}$$

Spam category (x)	Number of Students affect by spam	Probability
1	6	6/9 = 0.6667
2	1	1/9 = 0.1111
3	2	2/9 = 0.2222
4	0	0.0000
Total	9 students	1.0000



To calculate the probability for each events:
We Let random variable X = be number of spam categories received by the students.
Possible value x can be 1,2,3,4
The events that we would like to study is:

- The probability that student(from 30 students) is affected by spam email (9 students).

$$\mathrm{P(affected)} = \frac{\text{number of students affect by spam email}}{\text{total student}}$$

$$\mathrm{P(affected)} = \frac{9}{30} = 0.30 = 30\%$$

This show that a 30% of students are receives the same and also different spam emails.

- The probability that students is affected has exactly 1 spam email category.

DEFINITION

For any two events A and B with $P(B) > 0$, the **conditional probability of A given that B has occurred** is defined by

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad (2.3)$$

$$P(X = 1 \mid \text{affected}) = \frac{\text{number of students affect by spam email} \cap \text{student got 1 category}}{\text{total student affects by spam}}$$

$$P(X = 1 \mid \text{affected}) = \frac{6}{9} = 0.6667 = 66.67\%$$

This show that a 66.67% of students are receives the only one spam email category.

- The probability that the student has 2 or 3 spam emails categories.
  Student that has between 2 and 3 spam emails categories.

$$P(X = 2 \cup X = 3 \mid \text{affected}) = P(2 \leq X \leq 3 \mid \text{affected}) = \frac{\text{number of students affect by spam email} \cap \text{student got 2 or 3 category}}{\text{total student affects by spam}}$$

$$P(X = 2 \cup X = 3 \mid \text{affected}) = \frac{1}{9} + \frac{2}{9} = 0.1111 + 0.2222 = 0.3333 = 33.33\%$$

This show that a 33.33% of students are receives 2 or 3 spam emails categories.

- The probability that students receive Education spam.
  Base on data we have 4 students who receive Education spam.

$$P(\text{Education} \mid \text{affected}) = \frac{\text{number of students affect by spam email} \cap \text{student receive Education spam}}{\text{total student affects by spam}}$$

$$P(\text{Education} \mid \text{affected}) = \frac{4}{9} = 0.4444 = 44.44\%$$

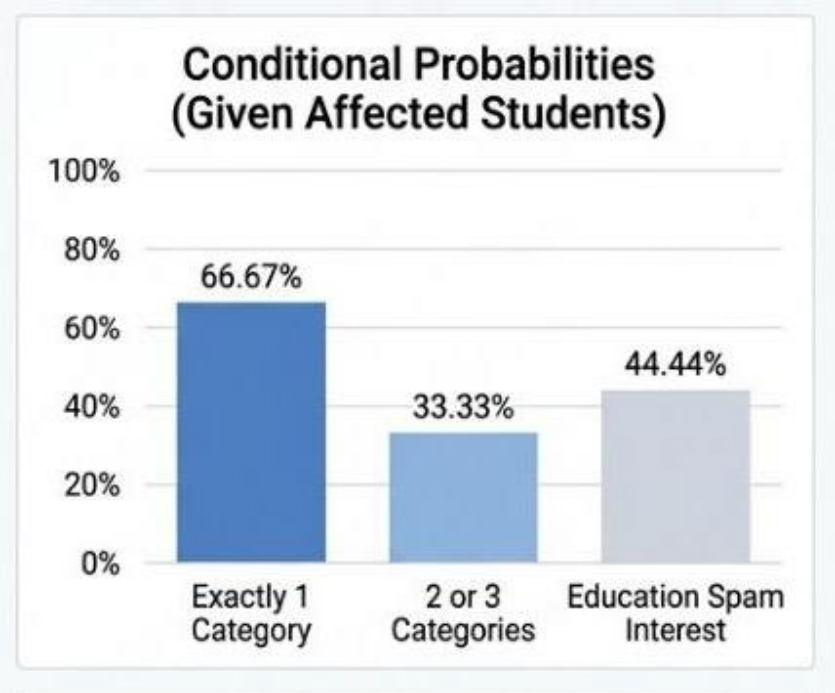
This show that a 44.44% of students are receives Education spam category.

- The probability that students receive AI / GetHub spam.
  Base on data we have 3 students who receive AI / GetHub spam.

$$P(\text{AI / GetHub } \mid \text{affected}) = \frac{\text{number of students affect by spam email } \cap \text{ student receive AI / GetHub spam}}{\text{total student affects by spam}}$$

$$P(\text{AI / GetHub } \mid \text{affected}) = \frac{3}{9} = 0.3333 = 33.33\%$$

This show that a 33.33% of students are receives AI / GetHub spam category.



2.5 Expected Value

We calculate the expected number of spam categories of affected student.

X is number of students who receive spam emails.

Spam category (x)	Number of Students affect by spam	Probability
1	6	6/9 = 0.6667
2	1	1/9 = 0.1111
3	2	2/9 = 0.2222
4	0	0.0000
Total	9 students	1.0000

The Expected Value of X

DEFINITION

Let X be a discrete rv with set of possible values D and pmf $p(x)$. The **expected value** or **mean value** of X, denoted by $E(X)$ or μ_X or just μ, is

$$E(X) = \mu_X = \sum_{x \in D} x \cdot p(x)$$

Expected value form by:

$$E(X) = \sum x \cdot P(X = x)$$

$$E(X) = 1 \cdot P(X = 1) + 2 \cdot P(X = 2) + 3 \cdot P(X = 3) + 4 \cdot P(X = 4)$$

$$E(X) = 1 \cdot \left(\frac{6}{9}\right) + 2 \cdot \left(\frac{1}{9}\right) + 3 \cdot \left(\frac{2}{9}\right) + 4 \cdot \left(\frac{0}{9}\right) = \frac{14}{9} \approx 1.5556$$

Among students who receive spam category emails, the average number or expected value is 1.5556, this show that we expect students might receive between 1 and 2 spam categories.

2.6 Variance and Standard Deviation

DEFINITION

Let X have pmf $p(x)$ and expected value μ. Then the **variance** of X, denoted by $V(X)$ or σ_X^2, or just σ^2, is

$$V(X) = \sum_D (x - \mu)^2 \cdot p(x) = E[(X - \mu)^2]$$

The **standard deviation** (SD) of X is

$$\sigma_X = \sqrt{\sigma_X^2}$$

A Shortcut Formula for σ^2

The number of arithmetic operations necessary to compute σ^2 can be reduced by using an alternative formula.

PROPOSITION

$$V(X) = \sigma^2 = \left[\sum_D x^2 \cdot p(x)\right] - \mu^2 = E(X^2) - [E(X)]^2$$

- Variance: $V(X) = E(X^2) - [E(X)]^2$

Find $E(X^2)$

$$E(X^2) = \sum x^2 \cdot P(X = x)$$

$$E(X^2) = 1^2 \cdot P(X = 1) + 2^2 \cdot P(X = 2) + 3^2 \cdot P(X = 3) + 4^2 \cdot P(X = 4)$$

$$E(X^2) = 1 \cdot \left(\frac{6}{9}\right) + 4 \cdot \left(\frac{1}{9}\right) + 9 \cdot \left(\frac{2}{9}\right) + 4 \cdot \left(\frac{0}{9}\right) = \frac{28}{9} \approx 3.1111$$

$$V(X) = 3.1111 - (1.5556)^2 \approx 0.6912$$

- Standard Deviation: $V(X) = \sigma_X^2$

Then $\sigma_X = \sqrt{V(X)} = \sqrt{0.6912} \approx 0.8313$

The variance (0.6912) and standard deviation (0.8313) indicate that the number of spam categories for affected students is relatively consistent.

Most affected students have 1 or 2 categories, falling within one standard deviation of the mean. Students with 3 categories are above average, while no student had 4 categories.

III. CONCLUSION

In this project, we analyzed the probability of spam email occurrences among ITC students through a survey of 30 participants. The study focused on the number of spam categories received by affected students, providing valuable insights into spam email patterns in an academic environment.

3.1 Main Result

The main results of the study are summarized in the table below which showing the key probability measures and their interpretations:

Measure	Value	Interpretation
Expected value	E(X) = 1.556	Average is about 2 categories
Variance	V(X) = 0.691	The spread among affected students is low
Stadard deviation	$\sigma_X = 0.831$	Number of spam categories is fairly consistent
P(X=1)	0.667	Most student got 1 category.
P(X=2)	0.111	Few student got 2 categories.
P(X=3)	0.222	Some student got 3 categories.

P(X=4)	0	No student got 4 categories.

The result shows that spam emails are quite common, affecting nearly 30% of students by spam, most of the affected students receive spam from only one category. The low standard deviation reflects relatively consistent spam behavior among affected users. Education and AI/GitHub spam were the most common categories observed.

3.2 Overall

Overall, spam emails have become a common issue for students in daily life. They waste our time, take up space in the inbox, and sometimes bring risks like scams or phishing. Our calculations found that 30% of students received spam in just two weeks. Among those who got spam, most only received one category and the average is about 1.56 categories. The small variance and standard deviation tell us that the number of spam categories is quite stable and predictable. Most students do not get spam from many different categories at once. These findings show that spam has a clear pattern. This information can help us build better spam filters and teach students how to protect themselves better when using email.

This project helped us apply important probability concepts like random variables, probability distribution, conditional probability, expected value, and variance to a real-life situation.

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